Midterm Part 1: Report

Start Assignment

- Due Tuesday by 5pm
- Points 20
- Submitting a file upload
- Available after Oct 12 at 2pm

Midterm Report 1: The Sensor

Individual assignment. You and your lab partner share a board, so you will report the same measured numbers. That is expected. The analysis, the explanations and the writing must be your own.

Everything here comes from **Part 1**. There is nothing to measure and no reason to go back to the bench.

20 points. Due: First 60 minutes of the session at the end of week 1, the same deadline as your Part 1 demonstration. Late submission costs 10% per hour.

Why this is due a week after you took the data, and not the same night

You finish the six orientations and the deliberately broken pair shortly before the Part 1 demonstration. Writing them up that evening would be the worst possible time.

You have the week instead, with the board still on the bench if you need to look at something again. Take the numbers carefully during the session and the writing is analysis rather than archaeology.

Question 1: The gravity table (10 points)

a. Include your six-orientation table: ax, ay, az and |a| for each.

b. Report the spread in |a| across the six orientations, and the mean.

c. Say which physical direction each sensor axis corresponds to on your board, and how you determined that from the data rather than assuming it.

d. The datasheet quotes a zero-g offset for this accelerometer. Find it, and work out how large a spread in |a| that offset alone could produce across six orientations. Compare with the spread you measured.

e. If your spread is substantially larger than the offset can account for, propose what else is contributing. Consider how level your bench is, how long you averaged, and whether the board was truly at rest.

f. Include your two deliberately broken orientations from Part 1c, and explain the signature.

Question 2: Diagnosis without the answer key (5 points)

Suppose a classmate shows you a six-orientation table and asks what is wrong with their code. For each of the following, say what single fault it points to and how you would confirm it.

a. All three axes report the same value in every orientation.

b. |a| is a constant 0.5 g in all six orientations.

c. |a| is correct with the board one way up and about 3 g when it is inverted.

d. |a| is correct in four orientations and wrong in the two where gravity is along the same axis.

e. The accelerometer has no identification register. Describe the known-answer test you built instead, and say what it proves that simply reading a register and judging whether the number looks sensible does not.

Question 5: The rate budget (5 points)

a. Report your sustained sample rate over the 30-second run, your SCL frequency and your bytes per transaction.

b. Account for the time. How much is the I²C transaction itself, and how much is everything else?

c. State how you read the six output bytes, and what your rate would have been with the other approach. Show the arithmetic rather than guessing.

d. Explain how you established that your samples are independent rather than repeated reads of the same conversion, and what the measured rate would look like if they were not.

e. If you needed ten times the rate, what would you change first, and what would stop you next?

Submission

Upload a single PDF, `midterm_report1.pdf`, containing your answers, the gravity table and any plots. Label the axes and give units on every number.

State clearly **which I²C state machine you used** — your hand-written one or your generated one — and what you had to change to talk to a different device on a different bus. Either is fine; not saying which is not.

Your **code goes to the Midterm – Part 1 assignment**, uploaded before your demonstration. Do not attach it here as well.